

14/7/26

TASK-5

* Compute the time complexity using the frequency count for the following sequence.

⇒ for (i=1; i ≤ 10; i=i+2)

{
 $\downarrow$ $\downarrow$ $\rightarrow 5$
 1 6

i = i + 10; — 5

printf ("i: %d", i); → 5

= 22

}

⇒ j=1; → 1

while (i ≤ 50); ~~100~~

= 1

{

printf ("hello"); ~~100~~

}

⇒ for (i=1; i ≤ 20; i++)

= 21

for (i=1; i ≤ n; i=i+2)

{
 $\downarrow$ $\downarrow$ $\rightarrow n/2 + 1$
 1 $n/2 + 1$

printf ("hello"); $n/2$

printf ("world"); $n/2$

= $\frac{5n}{2} + 2$

printf ("i = %d", i); $n/2$

}